

“ I NOW FELT... CERTAIN THAT
THE DAY WOULD COME WHEN
MANKIND WOULD BE ABLE TO
SEND **MESSAGES** WITHOUT WIRES
ACROSS THE ATLANTIC. ”

Guglielmo Marconi, Italian inventor, in *Messages without Wires*, 1901

The equipment at the Marconi wireless station in Massachusetts, US, which transmitted across the Atlantic to Cornwall, England.

Wright brothers experimented with glider design to maximize lift and added a lightweight aluminium gasoline engine to provide the power. **Orville made the first 12-second flight** in their aircraft, the 1903 Flyer, and flew 40yd (37m). The same day,



Orville and Wilbur Wright

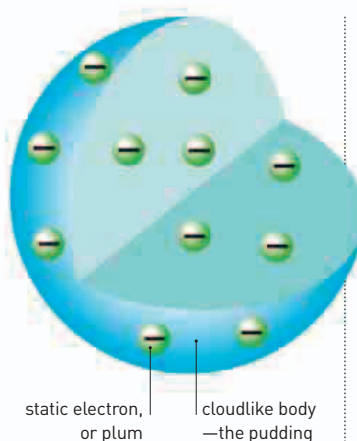
The Wright brothers ran a bicycle shop, but were inspired by early attempts at aviation. By 1908, they had managed an hour-long flight, and carried a passenger.

Wilbur managed 284yd (260m) in 59 seconds.

The Wright brother's first aircraft was designed to minimize load and maximize flexibility. It had a spruce ash wooden frame covered in muslin. The engine gave it enough speed for the wings to generate more lift than the weight of the machine: the **principle of flight**.

AFTER THREE YEARS of sending transatlantic radio signals—including coherent messages—Italian inventor **Guglielmo Marconi** (1874–1937) set up the **first commercial transatlantic radio service**. By 1907 it had become a regular service.

The March issue of the science journal the *Philosophical Magazine* contained a piece written by British physicist **J.J. Thomson** (1856–1940) in which he described a **new way of looking at the atom** that accommodated the newly discovered electron, known as the **plum pudding model**. Thomson thought of an atom as a “pudding” of positive charge, in which the negatively charged electron “plums” were embedded. Later in the year, the Japanese physicist **Hantaro Nagaoka** (1865–1950) **rejected the idea** that positive and negative charges could intermingle in this way, and suggested his **Saturnian model**.



The plum pudding model
Thomson's model for atomic structure was an early attempt at suggesting how charged particles could coexist in a neutral atom.

His model had a large positive core, which was orbited by negative electrons, rather like the rings around planet Saturn. Within a few years, experiments in the UK would show that atoms have a dense positive nucleus with encircling electrons—more like Nagaoka's model.

In July, Piero Conti (1865–1939), an Italian businessman from the volcanic region of Larderello, Tuscany, **demonstrated a steam engine that ran on geothermal power**. Conti succeeded in producing enough electricity from a dynamo to illuminate five light bulbs. Conti's legacy is that the Larderello region now produces 10 percent of the world's geothermal electricity.

In November, British physicist **John Ambrose Fleming** (1849–1945) **filed a patent for the first vacuum diode**—designed by adding a positive electrode (anode) to an Edison light bulb. Electrons from the hot filament bulb flowed through the bulb's vacuum to the cold anode—converting alternating current (AC) signals to direct current (DC). This marked the **start of an era of electronics**, and for decades improved versions of Fleming's diodes were used in many devices, from radios to the first computers.

In early December at the Lick Observatory in California,

Fleming's vacuum diode

The design features a metal plate that acts as an anode (positively charged electrode) to attract electrons from the bulb filament, thereby creating direct current.

American astronomer **Charles Dillon Perrine** (1867–1951) **discovered a moon of Jupiter** that was later **named Himalia**—after a mythical Greek nymph. It is the sixth largest of Jupiter's moons and is thought to have been formed from an asteroid captured in the planet's orbit.



7,000,000 MILES
THE DISTANCE OF HIMALIA FROM JUPITER

September 24
George Darwin suggests the Earth is heated by radioactivity

October 17 David Bruce reports that sleeping sickness is spread by tsetse flies

Dutch scientist Cornelius Koning suggests that fungi help decomposition of organic matter to form humus

March J.J. Thomson describes plum pudding model of atomic structure

June British astronomer **Edward Maunder** describes patterns of sunspot distribution

German biologist Friedrich Meves describes finding mitochondria in plant cells

July 4 Piero Conti tests the first generator powered by geothermal energy

November 16 British Patent Office registers the thermionic diode valve for John Fleming

December 5 Hantaro Nagaoka describes a Saturnian model of atomic structure

December 3 Dillon Perrine discovers the 10th moon of Jupiter